Progress Report 5

1. **Project title:**
   Touchscreen interface for a multipurpose robot (R1D4) (for the first semester) and swarm of robots to complete locomotive and object-searching tasks (for the second semester).
2. **Names, student numbers, and emails:**
   Group 36 members: (all members are registered in ECSE 458-D1 and ECSE 458-D2)

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3. **Project advisor(s):**
   Joseph Vybihal: joseph.vybihal@mcgill.ca

4. **Group meetings and meetings with advisor(s):**
   All four team members, along with the advisor, have agreed to hold two meetings each week: one with all team members present and another with the advisor and any available team members at that time.

   Weekly meeting with the entire group: Wednesdays at 3:00 pm (previously it was Tuesdays at 3:45pm)
   Weekly meeting with the advisor: Thursdays at 4:30 pm

   In addition to these scheduled meetings, this semester, we have decided to dedicate individual free time to working on the swarm of robots at Trottier 4130. This extra lab time will give us the opportunity to work directly with the hardware, identify and fix issues, and improve our system, helping us make steady progress outside of our regular meetings.

1. **Recent progress**

We have almost finished the integration for the R1D4 interface, with the ESP32, telnet and the frontend of the application. We have established communication from the telnet server over Wi-Fi to the ESP32 and then to the tablet, where the tablet or the application can receive data (in this case commands) from the ESP32. The communication also works the reverse order, that is, from the tablet, the application can send delivery data info, encoded as a string, to the ESP32 and is then relayed to the telnet server. This proof-of-concept provides the foundation for future

design teams to integrate the interface into the R1D4 robot when it is up and running. Documentation will be completed to help familiarize future design teams with our contributions.

As for the swarm robots, we are familiarising ourselves with the current state of the robots and their working. We have also started evaluating the swarm robots, by making a list of the functionalities and configurations of each robot, followed by the fixes that need to be done (if any).

2. **Future plans**

The next step in our plan is to get started on the Swarm Robots. We have become acquainted with them in the lab but have yet to get them to the same place they were before the end of last semester. After we get the swarm robots up and running, our first step will be to add the new robot to the mix. Then, we want to start using the sensor data that the robots are picking up, i.e. the ultrasonic data, to navigate. We are also looking into adding a camera that tracks the robot from an outside perspective and uses some sort of tracking method to help us guide the robot's movements. At this moment, the best idea we have is using specific colors for each robot and then finding them using colour thresholding techniques. We also are beginning to create our presentation for the mid-project demo. We will be demoing the finished R1-D4 frontend interface along with updating our project supervisor with the progress on the swarm robots.

3. **Professionalism**

Engineering plays a vital role in developing technology that is not only effective but also safe and reliable for public use. As we move into the second phase of our project, we are focusing on building a swarm of autonomous robots capable of performing locomotive and object-searching tasks. Since these robots will operate in real-world environments, ensuring safety, reliability, and efficiency is crucial. Any failure—whether it's a communication breakdown, navigation errors, or unintended collisions—could disrupt their ability to function as intended. To prevent this, we will implement rigorous testing protocols to assess each robot’s movement, task execution, and coordination with the central system. Our ultimate goal is to create a system that is not just operational but also predictable and safe, ensuring that the robots can function smoothly while minimizing risks in their environment.

To maintain high professional standards in the second phase of our project, we will prioritize thorough testing, modular design, and a structured development process. One of our main goals is to establish seamless communication between the swarm robots and the central control system. To achieve this, we need a reliable network that prevents disconnections and ensures each robot correctly receives and executes its assigned tasks. Additionally, we will implement fail-safe mechanisms so that if one robot encounters an issue, it can either continue operating

independently or adjust its tasks without disrupting the entire system. From a project management perspective, we will follow a structured workflow, including clear documentation, progress tracking, and continuous testing. This approach will help us refine our design at every stage, making sure the final system is not only technically sound but also meets ethical and engineering standards.

Our work on the R1D4 touchscreen interface in the first phase of the project gave us valuable insights that will improve our approach to the swarm robotics system. One of the biggest lessons we learned was the importance of good project management and clear task delegation. We saw firsthand how early communication and debugging helped prevent delays, so this time, we plan to start testing the hardware and software integration as early as possible in order to avoid last-minute issues. We also realized how crucial detailed documentation and open team communication were in solving problems efficiently. To keep things running smoothly, we will continue maintaining regular team updates, which will make sure that each member of the team stays on the same page. Another key change in our approach is how we think about usability and system design. While the R1D4 project focused on making the interface user-friendly, the swarm robots will require strong decision-making algorithms to coordinate effectively as a team. By applying what we learned in the first semester and maintaining high engineering standards, we aim to build a scalable, efficient, and reliable swarm robotics system that not only meets technical requirements but also serves the public safely and effectively.